

Multiple Regression: Mammal Size

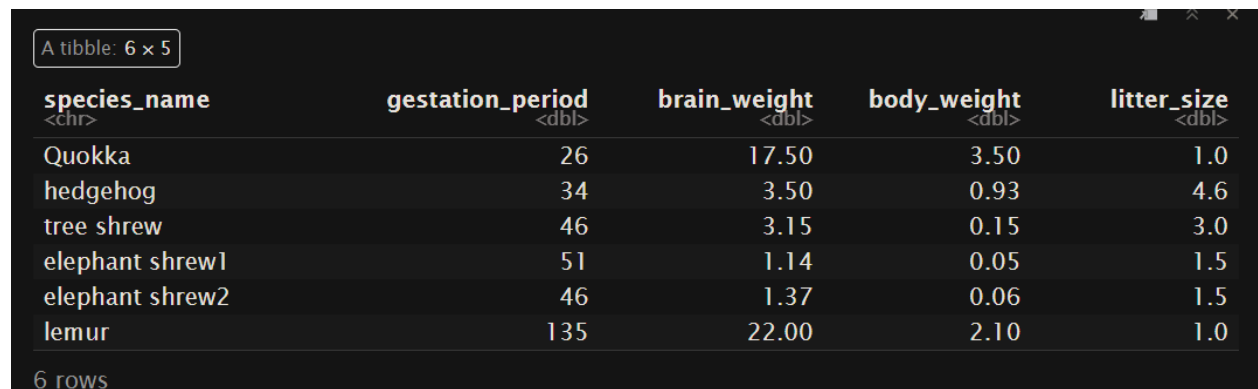
1. Data Description

The data consists of 96 records characterising various species of mammals.

The dataset explores factors influencing the gestation period, such as brain size, body weight, and litter size. It investigates possible trade-offs like longer pregnancies and fewer offspring for larger brain sizes. For each species attributes are provided:

Name of Variable	Variable Description
species_name	Name of the mammal species
gestation_period	Duration of pregnancy (in days)
brain_weight	Weight of the brain (in grams)
body_weight	Weight of the body (in kilograms)
litter_size	Number of offspring per birth

```
head(mammalsize)
```



A tibble: 6 × 5

species_name <chr>	gestation_period <dbl>	brain_weight <dbl>	body_weight <dbl>	litter_size <dbl>
Quokka	26	17.50	3.50	1.0
hedgehog	34	3.50	0.93	4.6
tree shrew	46	3.15	0.15	3.0
elephant shrew1	51	1.14	0.05	1.5
elephant shrew2	46	1.37	0.06	1.5
lemur	135	22.00	2.10	1.0

6 rows

```
summary(mammalsize)
```

```

species_name      gestation_period  brain_weight    body_weight
Length:96         Min.      : 16.0      Min.      :  0.45   Min.      :  0.020
Class :character   1st Qu.: 63.0      1st Qu.: 12.60   1st Qu.:  2.075
Mode  :character   Median :133.5     Median : 74.00   Median :  8.900
                  Mean   :151.3     Mean   : 218.98   Mean   : 108.328
                  3rd Qu.:226.2     3rd Qu.: 260.00  3rd Qu.:  94.750
                  Max.   :655.0     Max.   :4480.00   Max.   :2800.000

litter_size
Min.      :1.00
1st Qu.:1.00
Median :1.20
Mean   :2.31
3rd Qu.:3.20
Max.   :8.00

```

Summary gives us the minimum value, 1st quartile, median, mean, 3rd quartile and maximum value of each variable in the data.

```
str(mammalsize)
```

```

tibble [96 × 5] (S3: tbl_df/tbl/data.frame)
 $ species_name      : chr [1:96] "Quokka" "hedgehog" "tree shrew" "elephant shrew1" ...
 $ gestation_period: num [1:96] 26 34 46 51 46 135 90 135 139 180 ...
 $ brain_weight      : num [1:96] 17.5 3.5 3.15 1.14 1.37 22 12.8 9.9 54 73 ...
 $ body_weight       : num [1:96] 3.5 0.93 0.15 0.05 0.06 2.1 1.2 0.7 7.7 3.7 ...
 $ litter_size       : num [1:96] 1 4.6 3 1.5 1.5 1 1.2 1 1 1 ...

```

From the data structure, we can observe that the data consists of 96 observations and 5 variables. It also tells us the type of variable i.e. gestation_period, brain_weight, body_weight and litter_size are numerical variables and species_name is a categorical variable.

```

correlation = cor(mammalsize[,2:5])
correlation

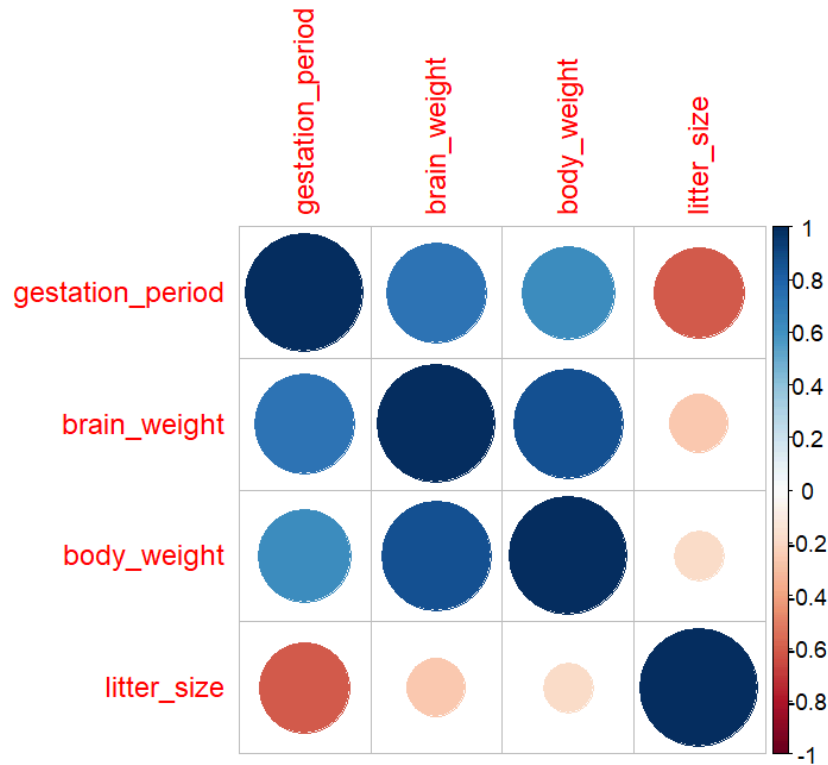
```

```

          gestation_period brain_weight body_weight litter_size
gestation_period      1.0000000      0.7281915      0.6246744     -0.6037515
brain_weight          0.7281915      1.0000000      0.8640269     -0.2545611
body_weight           0.6246744      0.8640269      1.0000000     -0.1827372
litter_size          -0.6037515     -0.2545611     -0.1827372      1.0000000

```

```
corrplot(correlation)
```



Above schedule and graph show the correlation between the variables. We can observe that brain weight and body weight are highly positively correlated with gestation period, which implies that as the brain weight or body weight of the animal increases, the gestation period tends to increase as well. However, litter size is highly negatively correlated with gestation period, indicating that species with longer gestation periods tend to have smaller litter sizes.

Additionally, there is a strong positive correlation between brain weight and body weight, which suggests that animals with larger brains tend to have larger bodies. However, litter size is negatively correlated with both brain weight and body weight, implying that species with larger brains and bodies tend to produce fewer offspring in a single litter.

2. Predictive Modelling

2.1 Multiple Regression

It is a statistical tool that allows you to examine how multiple independent variables are related to a dependent variable. Once we have identified how these multiple variables relate to our dependent variable, we can take information about all of the independent variables and use it to make much more powerful and accurate predictions about why things are the way they are. This latter process is called "Multiple Regression".

A population model for a multiple linear regression model that relates a y-variable to p-1 x-variables is written as

$$V_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{i(p-1)} + e_i.$$

Dependent Variable: Gestation Period (variable name: gestation_period)

Independent Variables:

Brain weight (variable name: brain_weight)

Body weight (variable name: body_weight)

Little size (variable name: litter_size)

Hypothesis:

We test,

$H_0: \beta_1 = \beta_2 = \beta_3 = \dots = \beta_p = 0$ i.e. there is no linear regression between our dependent and independent variable

VS

$H_1: \beta_i \neq 0, i=1,2,3 \dots p$, i.e. There is a linear regression between our dependent and independent variable

2.2 Model Building

Multiple linear regression model between our dependent and independent variables is

$$Y(\text{gestation_period}) = \beta_0 + \beta_1(\text{brain_weight}) + \beta_2(\text{body_weight}) + \beta_3(\text{litter_size})$$

Summary of Model

```
model <- lm(gestation_period ~ brain_weight + body_weight +
litter_size, data = mammalsize)
summary(model)
```

```

Call:
lm(formula = gestation_period ~ brain_weight + body_weight +
    litter_size, data = mammalsize)

Residuals:
    Min       1Q   Median       3Q      Max
-135.448  -38.675   -4.416   38.083  196.613

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  187.05056    11.04951   16.928 < 2e-16 ***
brain_weight    0.12197     0.02390    5.103 1.79e-06 ***
body_weight    0.01581     0.03618    0.437  0.663
litter_size   -27.79183     3.55161   -7.825 8.31e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 58.31 on 92 degrees of freedom
Multiple R-squared:  0.718,    Adjusted R-squared:  0.7088
F-statistic: 78.09 on 3 and 92 DF,  p-value: < 2.2e-16

```

2.3 Summary and Interpretation:

This is the summary output of a multiple linear regression model in R, where the dependent variable is `gestation_period` and the independent variables are `brain_weight`, `body_weight`, and `litter_size`.

1. F-test: (Test for Linearity)

Interpretation:

The F-test assesses whether the overall regression model is statistically significant. A very low p-value ($< 2.2e-16$) suggests that the model as a whole is highly significant and that at least one of the predictors (`brain_weight`, `body_weight`, `litter_size`) has a statistically significant relationship with the dependent variable (`gestation_period`).

2. t-tests for Individual Coefficients:

- `brain_weight`: Estimate = 0.12197, p-value = 1.79e-06 (significant)
- `body_weight`: Estimate = 0.01581, p-value = 0.663 (not significant)
- `litter_size`: Estimate = -27.79183, p-value = 8.31e-12 (significant)

Interpretation:

- The t-test checks whether each coefficient differs significantly from zero (no effect).
- brain_weight and litter_size have p-values well below 0.05, indicating they significantly affect the gestation_period.
- body_weight has a high p-value (0.663), indicating it is not significantly contributing to the model.

3. Multiple R²:

- $R^2 = 0.718$

Interpretation:

- Multiple R² measures the proportion of variance in the dependent variable (gestation_period) explained by the independent variables.
- 71.8% of the variation in gestation period is explained by the predictors included in the model.

4. Adjusted R²:

- Adjusted R² = 0.7088

Interpretation:

- Adjusted R² adjusts for the number of predictors in the model.
- It is slightly lower than R² (70.88%) and remains high, indicating a good model fit while considering the number of predictors.

5. Level of Significance:

Interpretation:

- *** indicates extremely significant predictors.
- Predictors with p-values < 0.001 (brain_weight, litter_size) are highly significant.
- body_weight is not significant at any conventional level.

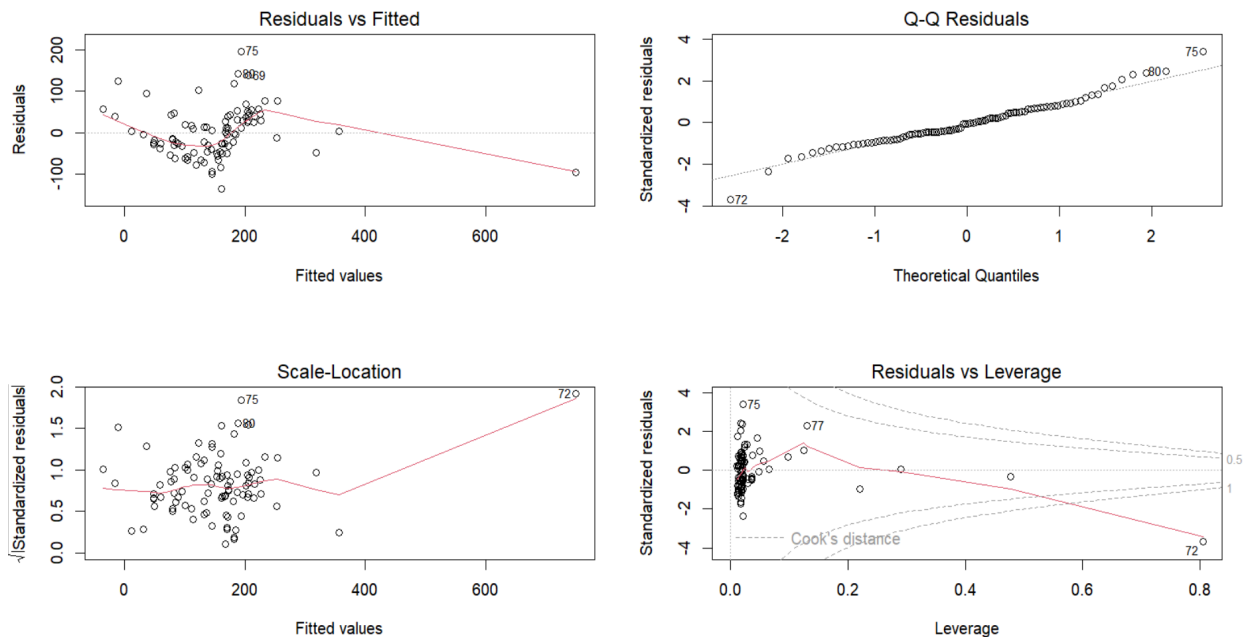
Overall Conclusion:

- The model is statistically significant overall (F-test result).
- brain_weight and litter_size are significant predictors of gestation period.
- body_weight is not a significant predictor.
- The model explains about 71.8% of the variation in gestation_period.

3. Assumptions Validation

3.1 Diagnostic Plots

```
par(mfrow=c(2,2))  
plot(model)
```



3.2 Diagnostic Tests

a) Multicollinearity: VIF Test

Multicollinearity exists when two or more of the predictors (independent variables) in a regression model are moderately or highly correlated. Unfortunately, when it exists, it can wreak havoc on our analysis and thereby limit the research conclusions we can draw. The VIFs of the linear regression indicate the degree that the variances in the regression estimates are increased due to multicollinearity. VIF values higher than 10 indicate that multicollinearity is a problem.

```
vif_test <- vif(model)  
print(vif_test)
```

brain_weight	body_weight	litter_size
4.101901	3.968616	1.075574

The VIF values do not exceed 10, hence we say that the model created using these variables is the best model and the problem of multicollinearity does not exist.

b) Homoscedasticity: Breush-Pagan Test

The assumption of homoscedasticity (meaning “same variance”) is central to linear regression models. Homoscedasticity describes a situation in which the error term is the same across all values of the independent variables.

Validation of homoscedasticity:

Homoscedasticity can be tested statistically by Breush-Pagan test.

Consider the hypothesis,

Null Hypothesis (H_0): The variance of the residuals is constant.

Alternative Hypothesis (H_1): The variance of the residuals is not constant.

```
bp_test <- bptest(model)
print(bp_test)
```

```
studentized Breusch-Pagan test

data:  model
BP = 1.7203, df = 3, p-value = 0.6324
```

This test has a p-value=0.6324 greater than a significance level of 0.05, therefore we do not reject the null hypothesis that the variance of the residuals is constant and infer that the assumption of homoscedasticity is satisfied. Alternatively, we can also detect homoscedasticity using the diagnostic plots.

c) Normality of Residuals: Shapiro-Wilk Test

```
shapiro_test <- shapiro.test(residuals(model))
print(shapiro_test)
```

```
Shapiro-wilk normality test

data:  residuals(model)
W = 0.97173, p-value = 0.03586
```

Null Hypothesis (H_0): The data follows a normal distribution.

Alternative Hypothesis (H_1): The data does not follow a normal distribution.

W = 0.97173, p-value = 0.03586 Since the p-value (0.03586) is less than 0.05, we reject the null hypothesis, indicating that the residuals are not normally distributed.

Remedial Measures:

Apply transformations to the dependent variable (log transformation)

d) **Autocorrelation: Durbin-Watson test**

```
dw_test <- durbinWatsonTest(model)
print(dw_test)
```

```
lag Autocorrelation D-W Statistic p-value
1      0.3562224      1.22879      0
Alternative hypothesis: rho != 0
```

Null Hypothesis (H_0): There is no first-order autocorrelation in the residuals.

Alternative Hypothesis (H_1): There is first-order autocorrelation in the residuals

D-W Statistic = 1.22879, p-value = 0

Since the p-value is 0, it indicates the presence of significant positive autocorrelation in the residuals.

Suggestion:

To address autocorrelation, Generalized Least Squares (GLS) modifies the regression model by weighting observations to account for correlated residuals, improving parameter estimates and standard errors.

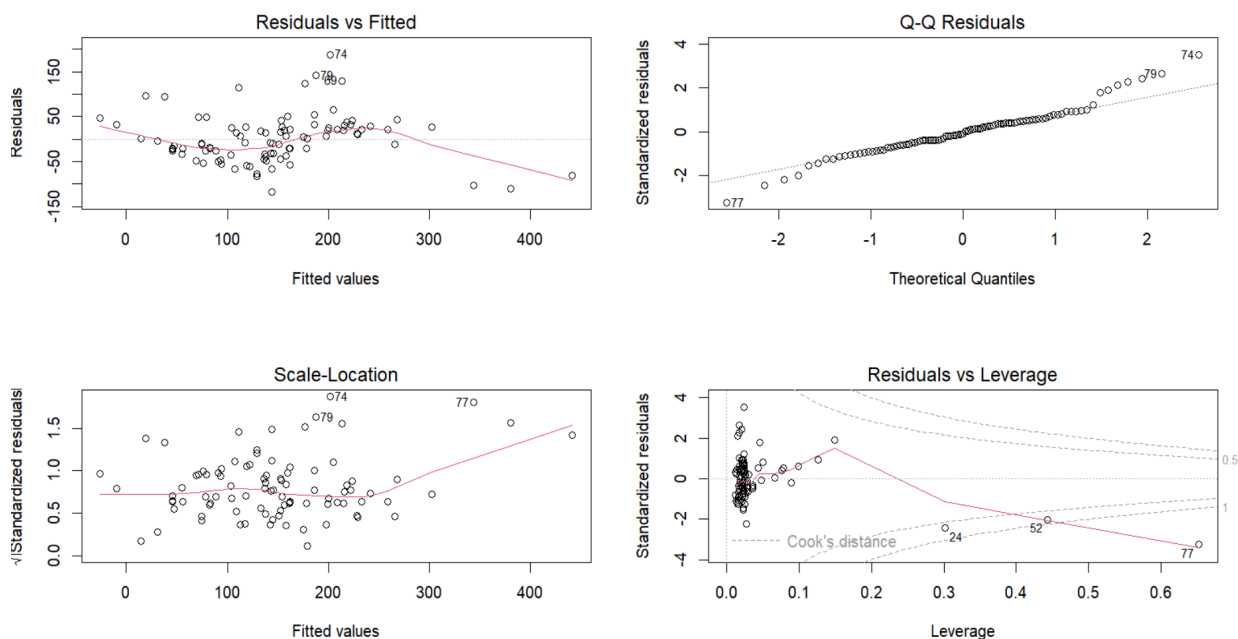
4. Removing Outliers

```
```{r}
mammalsize2 = mammalsize[-c(72),]
count(mammalsize2)
```
```

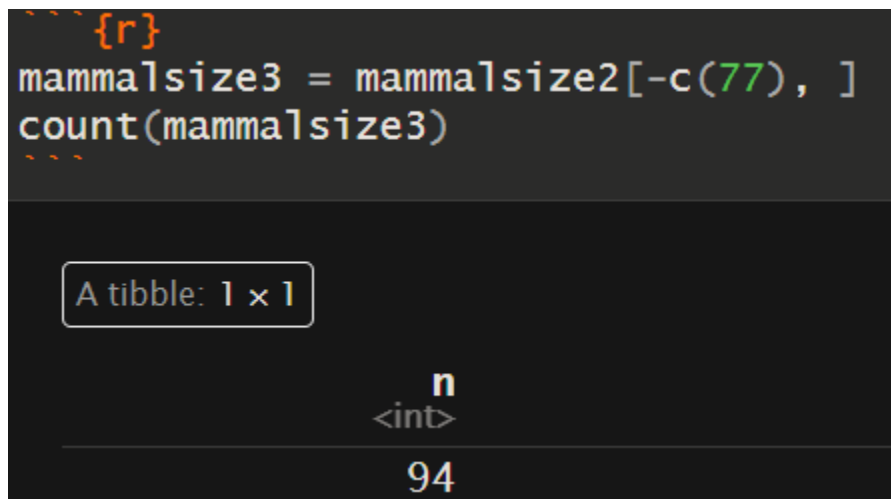
A tibble: 1 × 1

| n |
|-------|
| <int> |
| 95 |

Diagnostic Plots after removing observation 72



Final Count of the dataset after removing Outliers



The **Residuals vs Leverage** plot helps detect influential points and outliers. Ideally, residuals should be randomly scattered without any clear pattern, and no points should have both high leverage and high residuals. However, in this plot:

- ❖ Observations like **72**, **77** are labeled, indicating they may be influential points.

- ❖ Observation **72** appears to have high leverage with a relatively large residual, suggesting it may be influential.
- ❖ Cook's distance lines are also visible, showing the influence threshold and some points are near or beyond these lines.

Inference:

- ❖ There are influential points present in the dataset.
- ❖ The model may be affected by outliers, and adjustments or further investigation of these points might be necessary for a better model fit.

5. Fitting an OLS model after removing Outliers

```
Call:
lm(formula = gestation_period ~ brain_weight + body_weight +
    litter_size, data = mammals3)

Residuals:
    Min       1Q   Median       3Q      Max
-113.859  -30.466   -5.539   25.498  164.745

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  159.99879    11.16522   14.330 < 2e-16 ***
brain_weight    0.14784     0.02702    5.472 3.97e-07 ***
body_weight    0.22527     0.05681    3.966 0.000147 ***
litter_size   -23.51569     3.25570   -7.223 1.58e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 51.13 on 90 degrees of freedom
Multiple R-squared:  0.7212,    Adjusted R-squared:  0.7119
F-statistic: 77.61 on 3 and 90 DF,  p-value: < 2.2e-16
```

Adjusted R-squared: 71.19%, still indicating a good fit after adjusting for the number of predictors.

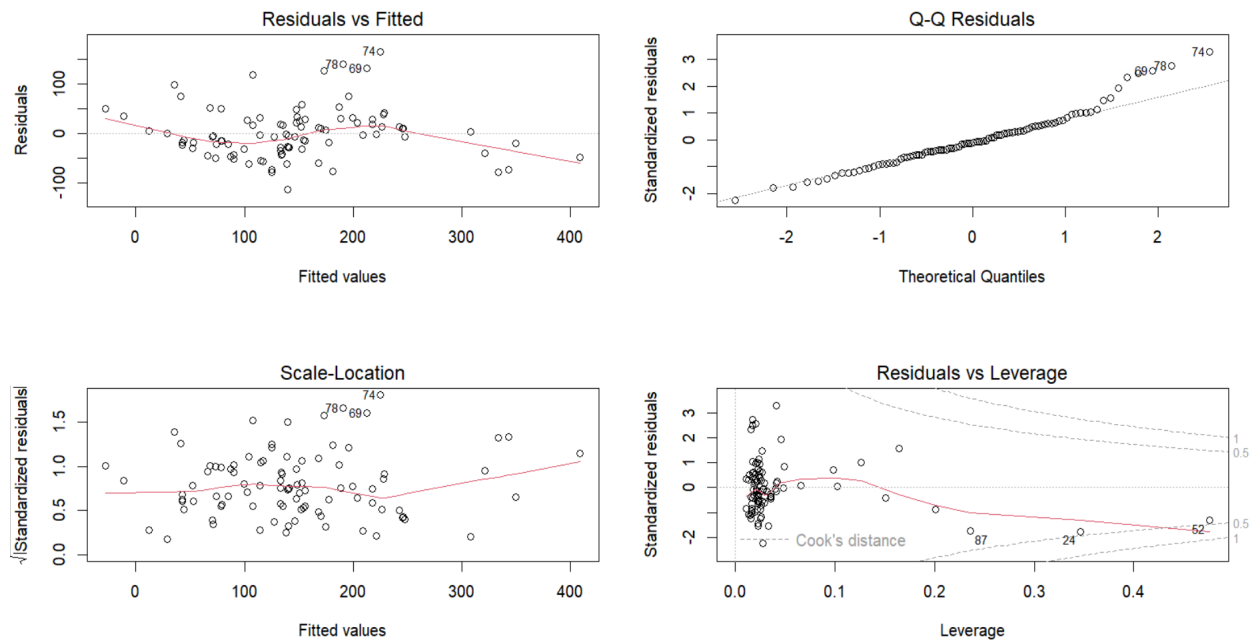
F-statistic: 77.61 with a p-value < 2.2e-16, suggesting the overall model is statistically significant.

T-test: All predictors are statistically significant, indicating they have a meaningful relationship with the gestation period.

5.1 Diagnostic Plots after removing Outliers

```
par(mfrow=c(2,2))
```

```
plot(model3)
```



Normality (Q-Q Plot): Very Slight deviations from the diagonal line suggest non-normality.

5.2 Assumptions Validation (using NICE Test)

```
nice_table(nice_assumptions(model3))
nice_assumptions(model3)
```

| Model | Normality
(Shapiro-Wilk) | Homoscedasticity
(Breusch-Pagan) | Autocorrelation of
residuals (Durbin-Watson) | Diagnostic |
|---------------------------|-----------------------------|-------------------------------------|---|------------|
| gestation_period ~ | | | | |
| brain_weight + | 0.00 | 0.64 | 0.00 | 2.00 |
| body_weight + litter_size | | | | |

The NICE test indicates:

- The **normality** and **autocorrelation** assumptions are violated (p-values < 0.05).
- The **homoscedasticity** assumption is satisfied (p-value > 0.05).

- The **diagnostic score** is 2 which indicates that the overall assumptions may need further exploration and improvement as we can see from the p-values itself.

Implications:

- The model's residuals are not normally distributed, which might impact inference (e.g., hypothesis tests or confidence intervals).
- Autocorrelation suggests that the residuals are correlated, which can lead to inefficient estimates.
- The homoscedasticity result indicates no evidence of unequal variance, so no corrective action is needed here.

Solution: To achieve the normality assumption we have conducted **log transformation** to the dependent variable of our data which has been shown below.

```
{r}
mammalsize3$t_gestation_period = log(mammalsize3$gestation_period)

model14 <- lm(t_gestation_period ~ brain_weight + body_weight + litter_size, data =
mammalsize3)
summary(model14)
```

```
Call:
lm(formula = t_gestation_period ~ brain_weight + body_weight +
    litter_size, data = mammalsize3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.57412 -0.27468  0.04969  0.24690  1.43848

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.0931897   0.1066969  47.735  < 2e-16 ***
brain_weight  0.0008376   0.0002582   3.244  0.00165 **
body_weight   0.0016517   0.0005428   3.043  0.00307 **
litter_size  -0.2814157   0.0311120  -9.045 2.77e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4886 on 90 degrees of freedom
Multiple R-squared:  0.6816,    Adjusted R-squared:  0.671
F-statistic: 64.22 on 3 and 90 DF,  p-value: < 2.2e-16
```

5.3 Summary and Interpretation:

This is the summary output of a multiple linear regression model in R, where the dependent variable is the transformed gestation period (`t_gestation_period`) and the independent variables are `brain_weight`, `body_weight`, and `litter_size`.

- **F-test:** p-value: $< 2.2\text{e-}16$

Interpretation:

The F-test assesses whether the overall regression model is statistically significant. A very low p-value ($< 2.2\text{e-}16$) indicates that the model as a whole is highly significant, meaning at least one of the predictors (brain_weight, body_weight, litter_size) has a statistically significant relationship with the dependent variable (t_gestation_period).

- **t-tests for Individual Coefficients:**

- brain_weight: Estimate = 0.0008376, p-value = 0.00165 (significant)
- body_weight: Estimate = 0.0016517, p-value = 0.00307 (significant)
- litter_size: Estimate = -0.2814157, p-value = $2.77\text{e-}14$ (significant)

Interpretation:

The t-test checks whether each coefficient is significantly different from zero (no effect). All three predictors (brain_weight, body_weight, litter_size) have p-values well below 0.05, indicating they significantly affect the transformed gestation period.

- **Multiple R^2 :**

- $R^2 = 0.6816$

Interpretation:

Multiple R^2 measures the proportion of variance in the dependent variable (t_gestation_period) explained by the independent variables. 68.16% of the variation in the transformed gestation period is explained by the predictors included in the model.

- **Adjusted R^2 :**

- Adjusted $R^2 = 0.671$

Interpretation:

Adjusted R^2 adjusts for the number of predictors in the model. It is slightly lower than R^2 (67.1%) and still indicates a reasonably good model fit while considering the number of predictors.

- **Level of Significance:**

Interpretation:

- *** indicates extremely significant predictors.
- All predictors (brain_weight, body_weight, and litter_size) are significant at conventional levels (p-values < 0.05 and 0.01).

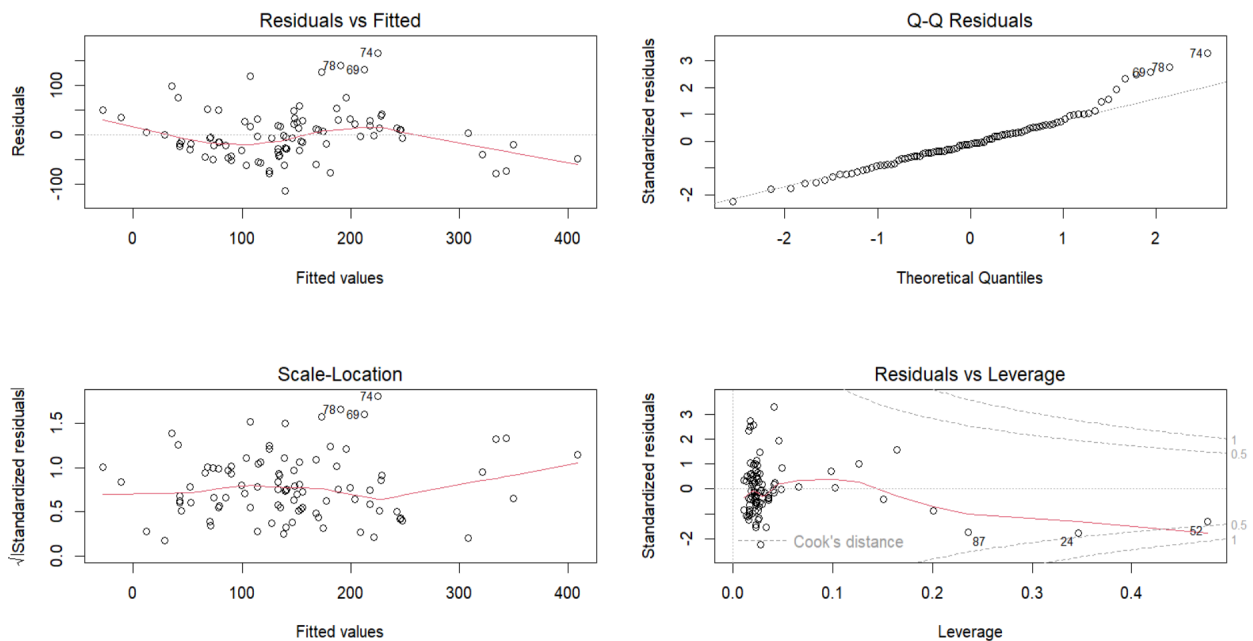
Overall Conclusion:

- The model is statistically significant overall (F-test result).
- All three predictors (brain_weight, body_weight, and litter_size) are significant predictors of the transformed gestation period.
- The model explains approximately 68.16% of the variation in the transformed gestation period.

5.4 Diagnostic Plots for for Transformed Variable Model

```
par(mfrow=c(2,2))
```

```
plot(model4)
```



5.5 Assumptions Validation for Transformed Variable Model

```

[1] "Variance Inflation Factor Test"
brain_weight  body_weight  litter_size
      1.645751      1.513511      1.161264

      Shapiro-Wilk normality test

data:  residuals(model4)
W = 0.9784, p-value = 0.1219

      studentized Breusch-Pagan test

data:  model4
BP = 8.5533, df = 3, p-value = 0.03586

lag Autocorrelation D-W Statistic p-value
  1          0.2316127          1.41581    0.008
Alternative hypothesis: rho != 0

```

Results of the Tests:

1. Variance Inflation Factor (VIF) Test: (To test Multicollinearity)

```

vif_test <- vif(model4)
print(vif_test)

```

Interpretation: VIF values below 5 (or sometimes 10) indicate that multicollinearity is not a significant concern. All three predictors have acceptable VIF values, suggesting minimal multicollinearity.

2. Shapiro-Wilk Normality Test

```

shapiro_test <- shapiro.test(residuals(model4))
print(shapiro_test)

```

Interpretation: Since the p-value (0.1219) is greater than 0.01, we fail to reject the null hypothesis that the residuals are normally distributed. This suggests that the residuals meet the normality assumption.

3. Studentized Breusch-Pagan Test

```

bp_test <- bptest(model4)
print(bp_test)

```


Interpretation: With a p-value of 0.03586 (more than 0.01), we do not reject the null hypothesis of homoscedasticity. This indicates the presence of homoscedasticity in the residuals.

4. Durbin-Watson Test

```
dw_test <- durbinWatsonTest(model4)
print(dw_test)
```

Interpretation: A Durbin-Watson statistic of 1.41581 indicates moderate positive autocorrelation. Since the p-value (0.008) is less than 0.05, we reject the null hypothesis of no autocorrelation. Autocorrelation is present.

6. Recommendations:

1. For Autocorrelation (Durbin Watson Test): At the 5% level of significance, the model exhibits autocorrelation. Thus we recommend fitting the model to a GLS (Generalised Least Squares) model. A GLS model is an extension of the ordinary least squares (OLS) regression that addresses violations of certain assumptions. GLS is used when the error terms are autocorrelated.

7. Conclusions

1. **Significance of Predictors:** Brain weight, body weight, and litter size significantly influence the gestation period.
2. **Model Fit:** The model explains approximately 71.8% of the variation in the gestation period ($R^2 = 0.718$), indicating a strong overall fit. After addressing outliers and transformations, the adjusted R^2 remains high.
3. **Assumption Validation:**
Autocorrelation persists in the residuals, which can impact the efficiency of estimates. A Generalized Least Squares (GLS) model is recommended to address this issue.
4. **Model Adjustments:** Removing outliers (e.g., observation 72) and applying a log transformation to the dependent variable (gestation period) improved normality.
5. **Real World Implications:** Larger brain and body sizes in mammals are associated with longer gestation periods but fewer offspring per litter, which may reflect evolutionary trade-offs. These insights have potential applications in understanding species' reproductive strategies and ecological adaptations.